Accelerated Gradient Methods

Optimization Methods, Week 4

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Why gradient descent can be slow

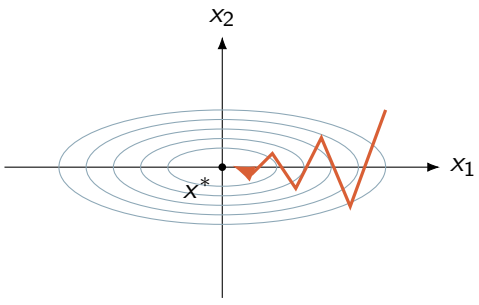
For $f(x) = \frac{1}{2}x^\top Qx - b^\top x$ with $Q \succ 0$,

$$e_{k+1} = (I - \alpha Q)e_k, \qquad e_k = x_k - x^*.$$

In an eigenbasis of Q, $z_{k+1,i} = (1 - \alpha\lambda_i)z_{k,i}$. One step must control every curvature scale. A step safe for $\lambda_{\max}$ makes progress of order $1/\kappa$ along the direction with $\lambda_{\min}$, where $\kappa = L/\mu$.

Geometry of an elongated quadratic

Level sets of $\frac{1}{2}x^\top Qx$ are ellipses. The gradient is normal to a level set, rather than pointing directly to the minimizer.



Large curvature causes oscillation across the valley. Small curvature controls progress along it.

The best constant step for a quadratic

Theorem

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The contraction is $\rho(\alpha) = \max_{\lambda \in [\mu, L]} |1 - \alpha\lambda|$. Because $|1 - \alpha\lambda|$ is convex in λ, the maximum occurs at an endpoint. At the optimum, the endpoint errors balance: $1 - \alpha\mu = -(1 - \alpha L)$. Solving gives $\alpha^* = 2/(L+\mu)$ and substitution gives $\rho^* = (L-\mu)/(L+\mu)$. □

Polyak momentum

Definition (Heavy-ball method)

With velocity $v_k = x_k - x_{k-1}$,

$$v_{k+1} = \beta v_k - \alpha \nabla f(x_k), \qquad x_{k+1} = x_k + v_{k+1},$$

where $\alpha > 0$ and $0 \leq \beta < 1$.

Momentum preserves motion along directions whose gradients agree over time. Alternating gradients across a narrow valley cancel part of the velocity.

Momentum as a second-order recurrence

For one eigen-direction with curvature λ,

$$z_{k+1} = (1 + \beta - \alpha\lambda)z_k - \beta z_{k-1}.$$

The characteristic polynomial is

$$r^2 - (1 + \beta - \alpha\lambda)r + \beta = 0.$$

Stability requires both roots to lie inside the unit disk for every $\lambda \in [\mu, L]$. The momentum coefficient controls memory and damping. Large β can accelerate a stable run or amplify oscillations when paired with a large learning rate.

Optimal heavy-ball parameters for quadratics

Theorem (Polyak)

For a positive definite quadratic with spectrum in $[\mu, L]$, the choices

$$\alpha^* = \frac{4}{(\sqrt{L} + \sqrt{\mu})^2}, \qquad \beta^* = \left(\frac{\sqrt{L} - \sqrt{\mu}}{\sqrt{L} + \sqrt{\mu}} \right)^2$$

give asymptotic factor $\rho = (\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$.

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Let $q = (\sqrt{L}-\sqrt{\mu})/(\sqrt{L}+\sqrt{\mu})$, so $\beta^* = q^2$.

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Limits of heavy-ball tuning

The optimal formula assumes an exact quadratic and known spectral bounds. For a general smooth strongly convex function, curvature varies with x and the same parameters need not give the quadratic guarantee.

- increasing β lengthens memory and reduces damping;
- increasing α strengthens both useful motion and overshoot;
- stability belongs to the pair (α, β), not to either parameter alone;
- restarts or adaptive schedules often improve robustness.

Coordinate descent

Suppose coordinate i has Lipschitz constant L_i:

$$|\nabla_i f(x + te_i) - \nabla_i f(x)| \leq L_i |t|.$$

A coordinate update is

$$x^+ = x - \frac{1}{L_i} \nabla_i f(x) e_i.$$

It costs one partial derivative and changes one variable. This is useful when gradients or updates are sparse, individual coordinates are cheap, or the dimension makes a full gradient expensive.

Decrease from one coordinate update

Theorem

Under coordinate smoothness,

$$f\left(x - \frac{\nabla_i f(x)}{L_i} e_i\right) \leq f(x) - \frac{(\nabla_i f(x))^2}{2L_i}.$$

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Coordinate selection rules

Cyclic Visit $1, 2, \ldots, n$ in a fixed order. It is simple and cache friendly, but the order can matter.

Randomized Draw i uniformly or with probabilities based on L_i. Randomization supports clean expected-rate analysis.

Greedy Choose the largest predicted decrease, for example $i = \arg\max_j |\nabla_j f(x)| / \sqrt{L_j}$. It often needs a costly scan.

The comparison must use work, rather than raw iteration count, because one coordinate step is cheaper than one full-gradient step.

Conjugate directions

For $Ax = b$ with $A \succ 0$, minimizing

$$q(x) = \tfrac{1}{2} x^\top A x - b^\top x$$

is equivalent to solving the system. Directions p_i, p_j are A-conjugate if

$$p_i^\top A p_j = 0 \quad (i \neq j).$$

Minimizing successively along mutually conjugate directions does not undo earlier minimizations.

The conjugate gradient algorithm

Initialize $r_0 = b - Ax_0$ and $p_0 = r_0$. For $k = 0, 1, \ldots,$

$$\alpha_k = \frac{r_k^\top r_k}{p_k^\top A p_k}, \qquad x_{k+1} = x_k + \alpha_k p_k,$$

$$r_{k+1} = r_k - \alpha_k A p_k, \qquad \beta_{k+1} = \frac{r_{k+1}^\top r_{k+1}}{r_k^\top r_k},$$

$$p_{k+1} = r_{k+1} + \beta_{k+1} p_k.$$

CG needs matrix-vector products with A, not a matrix factorization.

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$$\|x_k - x^*\|_A \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k \|x_0 - x^*\|_A .$$

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CG minimizes the A-norm error over $x_0 + \mathcal{K}_k(A, r_0)$.

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Finite-sum objectives and SGD

Machine learning often uses

$$f(w) = \frac{1}{N} \sum_{i=1}^{N} \ell_i(w).$$

A full gradient costs N sample gradients. SGD draws i_k or a mini-batch B_k and updates

$$w_{k+1} = w_k - \alpha_k g_k, qquad g_k = \frac{1}{|B_k|} \sum_{i \in B_k} \nabla \ell_i(w_k).$$

Under uniform sampling, $\mathbb{E}[g_k \mid w_k] = \nabla f(w_k)$.

Expected progress of SGD

Assume f is L-smooth, the estimator is unbiased, and $\mathbb{E}[\|g_k - \nabla f(w_k)\|^2 \mid w_k] \leq \sigma^2$.

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$$\mathbb{E}_k f(w_{k+1}) \le f(w_k) - \alpha_k(1 - L\alpha_k/2)\|\nabla f(w_k)\|^2 + \tfrac{L}{2}\alpha_k^2\sigma^2.$$

SGD rates and the noise floor

Under standard bounded-variance assumptions:

Setting	Step schedule	Representative rate
Convex	$\alpha_k \propto 1/\sqrt{k}$	$O(1/\sqrt{k})$ gap
Strongly convex	$\alpha_k \propto 1/k$	$O(1/k)$ gap
Smooth nonconvex	tuned constant	$O(1/\sqrt{k})$ average $\|\nabla f\|^2$

A constant step reaches a neighborhood whose size grows with $\alpha\sigma^2$. Larger batches reduce variance but cost more per update. Rates depend on the precise assumptions and averaging scheme.

Feature scaling as preconditioning

For least squares, $f(w) = \frac{1}{2N} \|Xw - y\|^2$ and

$$\nabla^2 f(w) = \frac{1}{N} X^\top X.$$

Large differences in feature scales create a large condition number. Standardizing columns changes the parameterization and often reduces spectral imbalance. This permits larger stable steps and shortens the zig-zagging phase. Correlated features can still leave the problem poorly conditioned.

Method selection

Method	Main advantage	Main requirement	Main risk
Momentum	cheap acceleration	tuned (α, β)	oscillation
Coordinate descent	cheap sparse updates	accessible partial gradients	misleading iteration comparisons
CG	fast quadratic solver	symmetric positive definite A	loss of conjugacy numerically
SGD	low cost per update	unbiased low-variance estimates	noise floor

Takeaways

- Poor conditioning explains slow progress and oscillatory trajectories.
- Polyak momentum can reduce the condition-number dependence on quadratics when its parameters are tuned well.
- Coordinate descent exploits cheap partial updates. CG exploits quadratic structure.
- SGD trades exact descent for inexpensive, noisy progress and requires a learning-rate schedule.
- Scaling changes optimization geometry before any optimizer runs.

Suggested reading: Polyak, *Introduction to Optimization*; Nocedal and Wright, *Numerical Optimization*; Bottou, Curtis, and Nocedal, "Optimization Methods for Large-Scale Machine Learning."